

The Intel Pentium IV processor from my first computer remains on my desk not as a relic, but as the catalyst for my academic trajectory. To my younger self, booting up that machine felt akin to starting a car, with the microprocessor serving as its engine. I was profoundly captivated by how invisible electrical currents could be orchestrated to render vibrant videos and immersive environments, igniting a childhood ambition to one day architect my own processor. What began as an early fascination with these inner mechanics has since matured into a rigorous academic pursuit, driving my commitment to designing efficient computing systems from the transistor level upward.

This curiosity, paired with my interest in mathematics, shaped the path I took. I earned a national rank of 149 out of 19,508 candidates at the G.C.E. Advanced Level Examination, which secured my admission to the **Faculty of Engineering at the University of Peradeniya**. I graduated **first in a class of 486** with a **B.Sc. Engineering (Honours) in Computer Engineering** and a perfect **GPA of 4.00/4.00**. Throughout my degree, I operated as a proactive learner, independently investigating concepts well before they were introduced in the classroom. I was less interested in the marks themselves than in figuring out how systems actually behave from the inside, and how minute architectural decisions can yield immense differences in performance.

Computer Architecture, which I encountered in my second year, was the subject that pulled everything together for me. It gave me a clear picture of how processors, caches, and memory hierarchies cooperate to make a system efficient. When elective time came, I picked Advanced Computer Architecture, Advanced Embedded Systems, Image Processing, Neural Networks, and Artificial Intelligence, partly because I wanted the hardware foundation to go deeper, and partly because I was curious about where intelligence and silicon meet.

While building and simulating a single-cycle processor in Verilog for my Computer Architecture course introduced me to hardware design, my first real applied research experience came from a JPEG encoding project in Advanced Embedded Systems. We built a **pipelined multiprocessor system-on-chip on an Altera Cyclone IV FPGA using Nios II cores**. The pipeline ran in software at first, and after simulating it, we found that several stages, especially the discrete cosine transformation, were stalling the processors. I added lightweight custom hardware instructions and FIFO queues to offload the heavy stages, which gave us a **2.8× throughput improvement**, and the work was published at [IEEE ICAC 2024](#). More importantly, the project taught me how I like to work: profile, find the bottleneck, and fix it at the architectural level. That instinct has shaped everything I have done since.

My final-year project, **SNAP-V**, grew out of that same instinct. With my team, I designed a **RISC-V based system-on-chip** with configurable **neuromorphic acceleration**, aimed at **small-scale spiking neural networks** for edge and robotics workloads. The goal was to take brain-inspired computing out of theory and into something compact and power-efficient enough to be useful. I worked across the full digital design flow, leveraging the Synopsys toolchain for RTL simulation and synthesis. Ultimately, we

validated the architecture in 45nm CMOS, achieving an average synaptic energy of **1.05 pJ per synaptic operation** with only a **2.62% hardware-software accuracy deviation** across 80 network configurations on MNIST. The work, supervised by Prof. Roshan G. Ragel and Dr. Isuru Nawinne, was formally presented as my [undergraduate thesis](#) and has produced two manuscripts, a core [methodology paper](#) and a comprehensive [review paper](#), now under peer review, with preprints available. I am currently extending it by deploying the accelerator on a **Xilinx Virtex-7 FPGA** and building a PCIe-based host interface so inference can be driven from a C runtime. Alongside this, I am currently continuing this research under the [PeraMorphIQ Neuromorphic Research Group](#), where I lead and mentor junior research teams working on on-chip learning integration and power optimization for next-generation neuromorphic accelerators. As a next step in my own work, I am actively exploring in-memory computing to enable low-power, memory-centric neuromorphic systems, while simultaneously investigating novel edge use cases for the SNAP-V platform.

To sharpen my microarchitectural foundation further, I designed and verified a five-stage pipelined [RISC-V RV32IM processor](#) with in-order hazard handling and prototyped it on a Xilinx Virtex-7 FPGA, reaching **142 MHz at 0.197 mW**. I am now extending this pipeline with a spiking neural network-based branch predictor, exploring whether biologically inspired techniques can improve prediction accuracy and energy efficiency. The work sits squarely at the intersection of conventional pipelines and event-driven computation, exactly the space I find most interesting.

Outside of research, my internship at [WSO2](#) gave me exposure to large-scale system design through Ballerina integrations, including the OpenAI Finetunes Connector and ISO20022-to-SwiftMT modules. I now work part-time at [Attune Integration](#) on hardware-software co-design for a commercial robotic system, which keeps me close to real deployment constraints. I also serve as a [Teaching Assistant](#) in the Department of Computer Engineering, running Computer Architecture and Embedded Systems labs and mentoring project teams.

Looking ahead, my research ambitions are firmly rooted in hardware-software co-design. Guided by my personal motto to "**make things unique**," my drive to continuously learn and push boundaries is focused on building systems that break from convention. I want to explore how unconventional architectural optimizations and custom accelerators can break through performance barriers across a wide variety of computational domains. I am intrinsically driven by the challenge of identifying systemic bottlenecks and translating abstract theoretical concepts into tangible, high-performance silicon realities. The journey from booting up that sluggish Pentium IV to architecting the next generation of uniquely optimized processors is a demanding one, but it is the exact path I am committed to walking.